Accessibility Repair Calculus

Local R2 Repair and Exceptional Loci for Length-2 Witnesses

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This paper is Part V of the RIME program. Paper III gives the first Rubik-centered accessibility separation example: composition-generated accessibility can exceed Lie-generated accessibility. The present paper isolates the stable static core behind that phenomenon: binary support is insufficient, projected commutator survival is the first repair layer, and length-2 accessibility witnesses admit a local repair calculus.

Abstract

Problem. Once sectors are fixed, how is accessibility computed? A sectorized representation supplies projectors Q_i and skew-Hermitian generators X_g, but the binary support graph is too coarse: it sees possible paths and forgets whether projected products survive, cancel, or vanish by incidence.

Approach. We separate support from relation data. The first layer R_1 records generator support, $Q_i X_g Q_j \neq 0$. The second layer R_2 records projected commutator survival, $Q_i[X_g, X_h]Q_j \neq 0$. Both are organized inside a weighted Hall path algebra with edge weights $W(i, g, j) = Q_i X_g Q_j$.

Results. The binary implication $(G, \chi, \Lambda) \to D$ is false. A three-generator S_4 example exhibits path-commutator cancellation: the binary graph predicts depth 1, but the projected commutator vanishes and the first nonzero Lie block appears at depth 2. For pairs with length-2 R_1 witnesses, the local behavior falls into four mechanisms: singleton-color degeneracy, commutator survival, Type III cancellation, and Type IV incidence. Type III is relation-level signed cancellation; Type IV is product-level image-kernel incidence. In the support-level obstruction calculus, R_2 repairs valid bridge failures through single-term commutator bridges forced by incomparability. For matrix blocks, this repair requires actual R_2 survival data, rank protection, or another nondegeneracy certificate.

Implications. Accessibility is not a binary graph invariant. It is a weighted Hall path problem, with D obtained as a valuation of Hall coefficients. The stable conclusion is a local one: R_1 is insufficient, R_2 is the first structural repair layer beyond binary support, and length-2 witnesses admit a closed local obstruction calculus. The later completion perspective upgrades the role of Type III and Type IV: Type III is the soft cancellation component, while Type IV is the hard incidence component. This Type I–IV vocabulary is a mechanism taxonomy, distinct from the moduli-wall hierarchy developed later in Paper VI. Full $(R_1, R_2) \to D$ completeness remains a theorem program rather than a closed theorem.

Notation Table

Symbol	Meaning
V	finite-dimensional complex Hilbert space
Q_i	orthogonal sector projector, $\sum_i Q_i = I$
X_g	skew-Hermitian generator, usually $X_g = \log \rho(g)$
$W(i, g, j)$	projected generator block $Q_i X_g Q_j$
$R_1(i, j; g)$	generator support: 1 iff $Q_i X_g Q_j \neq 0$
(G, χ, Λ)	binary colored accessibility graph encoded by R_1
$R_2(i, j; g, h)$	projected commutator survival: 1 iff $Q_i [X_g, X_h] Q_j \neq 0$
B_{ij}	infinite-depth channel discriminator $Q_i M_d(\mathbb{C}) Q_j$ in a block model
H_{ij}	Hall coefficient object: projected Hall monomials and their norms
D_{ij}	first nonzero Lie-depth from sector j to sector i
Sig	accessibility signature, the histogram of entries of D
Type I	singleton-generator length-2 path: $R_2 = 0$ locally
Type II	at least one projected commutator survives: $R_2 = 1$
Type III	cancellation locus: nonzero products survive termwise but cancel in projected commutators
Type IV	incidence locus: candidate products vanish termwise by $AB = 0$

Depth convention: depth 0 means a single generator block $Q_i X_g Q_j$. Depth 1 means a commutator block $Q_i [X_g, X_h] Q_j$. Depth 2 means a nested commutator such as $Q_i [X_a, [X_b, X_c]] Q_j$.

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1 Introduction

How is accessibility computed once sectors exist? The answer is not binary support alone. Accessibility is computed from a weighted Hall path algebra: the blocks $Q_i X_g Q_j$ give possible moves, while signed products of those blocks determine whether Lie brackets actually survive after projection.

Paper III proves a separation phenomenon in the Rubik cubie representation. There are sector pairs that are reachable by finite composition but invisible to the Lie algebra generated by the same representation. This raises the static problem studied here: given sector projectors and generators, determine the first Lie depth at which a sector pair becomes visible.

A tempting simplification is the binary colored graph:

its vertices are sectors, and it has an edge of color g from j to i precisely when $Q_i X_g Q_j \neq 0$.

This graph records where each generator has nonzero projected support. It is the natural first abstraction because it ignores metric data and remembers only the possible sector-to-sector moves.

The central correction of this paper is that binary support is not enough.

The graph sees that two colored paths exist. It does not know whether their projected matrix products cancel in a Lie bracket. Thus first depth is not determined by support alone.

The corrected object is a weighted Hall path algebra. Its edge weights are the blocks

$$W(i, g, j) = Q_i X_g Q_j.$$

The binary graph is only the support of these weights. Lie brackets are formed by signed sums of path products. Whether a bracket survives after projection depends on those products, not only on their nonzero pattern.

The first repair layer is R_2:

$$R_2(i, j; g, h) = 1 \iff Q_i [X_g, X_h] Q_j \neq 0.$$

This paper organizes the current state around a deliberately local target:

1. R_1 records generator support.
2. R_1 is insufficient.
3. R_2 records projected commutator survival.
4. Length-2 R_1 witnesses admit a four-type R_2 obstruction classification, with Type III and Type IV separated as distinct exceptional mechanisms.
5. In the support-level obstruction calculus, R_2 repairs valid bridge failures.
6. Class-level accessibility completeness still requires a global length-2 witness reduction and matrix nondegeneracy.

The result is not a universal depth theorem. It is a closed length-2 obstruction calculus plus a precise theorem program for higher Hall layers.

Remark 1.1 (Two Classification Axes).

The Type I–IV terminology in this paper is a **static local mechanism classification**. It applies after a sectorized system and a sector pair have been fixed, and it asks why a length-2 R_1 witness does or does not survive at the projected commutator layer R_2.

This is a different axis from the wall hierarchy used in Paper VI. Paper VI studies moving sectorized systems on normal spectral charts $\Sigma_{\mathrm{spec}} \subseteq \Sigma_{\mathrm{comm}}$ and classifies loci where the discrete observables $R_1(w)$, $R_2(w)$, or $D(w)$ change:

$$\Sigma_{R_1}, \qquad \Sigma^{\circ}_{R_2}, \qquad \Sigma^{\circ}_{D}.$$

Thus the same symbols R_1 and R_2 appear in both papers, but the classification axes are not the same:

Paper	Axis	Object classified
Paper V	length-2 obstruction mechanism	Type I–IV behavior of a fixed sector pair
Paper VI	moduli wall hierarchy	jumps of $R_1(w)$, $R_2(w)$, or $D(w)$ on normal spectral charts inside Σ_{comm}

In this dictionary, Paper V's Type III cancellation and Type IV incidence are two concrete local mechanisms that can appear inside a Paper VI $\Sigma^{\circ}_{R_2}$ wall: R_1 is held fixed, but projected commutator survival changes.

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2 Claim Status Summary

Claim	Status in this paper
Binary support (G, χ, Λ) does not determine D in the local realization model	established by path-commutator cancellation plus support-preserving perturbation
R_1 admits complement and non-complement bridge obstructions	support-level exhaustive evidence and structural analysis
R_2 is not derived from R_1	definition-level distinction; required by cancellation
Type I–IV classification covers pairs with length-2 R_1 witnesses	Theorem 7.1; local classification
Single-term bridge mechanism repairs valid centered length-2 support obstructions	Proposition 8.3; support/scalar level with stated support restriction
Type III failures are signed-product cancellation loci	Theorem 7.1 and Proposition 8.5; local exceptional locus
Type IV failures are image-kernel incidence loci	Theorem 7.1, Proposition 8.5, and Lemma 9.1; local exceptional locus
Matrix-block single-term products survive under rank protection	lemma; S4-3gen-B verified example
Matrix-level local repair beyond rank-protected cases	open upgrade / conditional program
Type IV incidence has high codimension in matrix-pair space	algebraic completion input; not needed for local classification
D is observed to be constant on exact (R_1, R_2) signature classes in tested structured systems	observed computational evidence for the later completion program; see the completion-atlas audit
Every finite-depth accessible pair has a length-2 R_1 witness	open Layer C reduction
Universal arbitrary-quiver $N = 2$ completeness	not claimed; sparse random counterevidence exists
Represented/dense $(R_1, R_2) \to D$ completeness	program-level conjecture

This table is part of the statement discipline of the paper. The proved local mechanisms are separated from the full Accessibility Completeness program. The last row is a static/global completeness question: it asks whether D can be recovered from (R_1, R_2) throughout a represented or dense class of sectorized systems. It is not changed by Paper VI's local stratification theorem, which organizes where $R_1(w)$, $R_2(w)$, and $D(w)$ jump on normal spectral charts inside the commutative moduli wall.

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3 Static Sectorized Observable Framework

Definition 3.1 (Sectorized Observable Framework)**.**
A static sectorized observable framework consists of:

1. a finite-dimensional complex Hilbert space V;
2. a finite orthogonal projector decomposition

$$I = \sum_i Q_i, \qquad Q_i Q_j = \delta_{ij} Q_i;$$

3. a finite family of skew-Hermitian generators

$$\mathcal{X} = \{X_g : g \in S\}.$$

In representation-derived examples, X_g is typically the skew-Hermitian logarithm of a unitary representation matrix:

$$X_g = \log \rho(g).$$

The projectors Q_i may come from a spectral center, a joint eigenspace decomposition, or another chosen sectorization. The finite-group representation background for these examples follows the standard semisimple representation setting [12].

Definition 3.2 (Lie-Depth Matrix)**.**
Let $\mathcal{L}^{(0)}$ be the span of the generators X_g. For $d \geq 1$, let $\mathcal{L}^{(d)}$ be the span of Lie monomials of bracket depth d in the X_g.

For sector pair (i, j), define

$$D_{ij} = \min\{d \geq 0 : Q_i Y Q_j \neq 0 \text{ for some } Y \in \mathcal{L}^{(d)}\},$$

with $D_{ij} = \infty$ if no such d exists.

The accessibility signature is a compressed histogram of D:

$$\mathrm{Sig} = (A_0, A_1, A_2, A_\infty),$$

where A_d counts pairs first visible at depth d and A_∞ counts frozen pairs. In systems where higher depths occur, the signature can be extended.

Definition 3.3 (Projected Edge Weights)**.**
The weighted sector edge associated to generator g is

$$W(i, g, j) = Q_i X_g Q_j.$$

The binary support layer is

$$R_1(i, j; g) = 1 \quad \Longleftrightarrow \quad W(i, g, j) \neq 0.$$

Equivalently, R_1 is a colored directed graph with vertex set the sectors and edge colors the generators.

Definition 3.4 (Projected Commutator Survival)**.**
The second relation layer is

$$R_2(i, j; g, h) = 1 \quad \Longleftrightarrow \quad Q_i [X_g, X_h] Q_j \neq 0.$$

Expanding through the sector resolution gives

$$Q_i [X_g, X_h] Q_j = \sum_k \left(Q_i X_g Q_k X_h Q_j - Q_i X_h Q_k X_g Q_j \right).$$

Thus R_2 is the survival status of the full projected commutator after summing all intermediate sector contributions. It is not a separate equality test at each intermediate sector, and it is not determined by R_1.

4 Weighted Hall Path Algebra

The free Lie algebra on the generators has a Hall basis: generators, commutators, nested commutators, and so on [6, 11]. Projecting a Hall monomial between sectors gives a signed sum of colored path products.

For example:

$$Q_i[X_g, X_h]Q_j = \sum_k \left(W(i,g,k)W(k,h,j) - W(i,h,k)W(k,g,j)\right).$$

At higher depth, a Hall tree produces a larger signed sum over colored sector paths. The Hall coefficient object records these projected monomials:

$$H_{ij} = \{T \mapsto Q_i B_T Q_j\},$$

where T ranges over Hall trees and B_T is the corresponding Lie monomial.

The depth matrix is the valuation of H:

$$D_{ij} = \min\{\mathrm{depth}(T) : Q_i B_T Q_j \neq 0\}.$$

Thus the correct relation is not

$$\text{binary graph} \longrightarrow D.$$

It is

$$\text{weighted Hall coefficients} \longrightarrow D.$$

The central compression problem is to determine how much of the weighted data is needed to recover D.

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5 Binary Support Is Insufficient

Proposition 5.1 (Binary Insufficiency in the Local Realization Model)**.**

The binary colored graph (G, χ, Λ) determined by R_1 does not determine the first depth matrix D in the local matrix-realization model.

Reason. *Binary support records only whether each projected block is zero or nonzero. It does not record whether two nonzero path products are equal, collinear, or otherwise cancel after projection.*

The minimal observed mechanism is path-commutator cancellation. The binary graph sees two length-2 paths

$$i \xrightarrow{g} k \xrightarrow{h} j, \qquad i \xrightarrow{h} k \xrightarrow{g} j,$$

but the projected products satisfy

$$Q_i X_g Q_k X_h Q_j = Q_i X_h Q_k X_g Q_j.$$

Therefore

$$Q_i[X_g, X_h]Q_j = 0,$$

even though the binary graph predicts a depth-1 commutator channel. This vanishing is a polynomial equality among the allowed block entries, not a condition encoded by the support *pattern itself. A support-preserving perturbation of one allowed block generically breaks the equality while leaving (G, χ, Λ) unchanged. Thus two local realizations with the same binary support can have different first-depth behavior.*

5.1 S4 Path-Commutator Cancellation

The current canonical counterexample is a three-generator S_4 system from the exploratory atlas. For one sector pair, the binary prediction is depth 1, while the observed first depth is depth 2.

The binary graph sees compatible length-2 paths, but every depth-1 projected commutator vanishes. The first nonzero Hall projection appears at the next nested depth.

This example is the concrete reason the unconditional implication

$$(G, \chi, \Lambda) \to D$$

is false.

Remark 5.2 (Why Earlier Binary Tests Passed).

Earlier two-generator tests passed because the relevant depth-2 leaf multisets often had a single Hall tree. With no same-multiset multi-tree cancellation available, the binary graph appeared sufficient. The three-generator counterexample exposes the missing relation data.

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6 R1 Failure as a Structural Problem

Binary insufficiency is not only a numerical cancellation issue. The independent R_1 support model also admits structural bridge obstructions.

Fix two boundary sectors, denoted 0 and 1, and a set V' of intermediate vertices. For each generator g, define the source-side and sink-side supports:

$$A_g = \{u \in V' : Q_u X_g Q_0 \neq 0\}, \qquad B_g = \{u \in V' : Q_1 X_g Q_u \neq 0\}.$$

The obstruction pattern is:

$$A_g \cap B_g = \emptyset, \qquad A_g \cap B_h \neq \emptyset \quad (g \neq h).$$

Each generator fails to close its own diagonal, but different generators cross.

6.1 Complement Family

The complement family is

$$A_g = \{u_g\}, \qquad B_g = V' \setminus \{u_g\},$$

with the u_g distinct. Then

$$A_g \cap B_g = \emptyset, \qquad A_g \cap B_h \neq \emptyset \quad (g \neq h).$$

Exhaustive support searches found stable R_1 survivors of this type:

Universe	Survivors
$\lvert V' \rvert = 3$	8/8
$\lvert V' \rvert = 4$	512/512

Per-generator forward closure does not destroy these obstructions. This is the structural failure of R_1: independent generator supports can satisfy all off-diagonal crossing conditions while keeping every diagonal empty.

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7 Local R2 Obstruction Classification

The following classification applies to sector pairs (v, w) satisfying:

1. $R_1(v, w) = 0$;
2. there exists at least one length-2 R_1 path $v \to k \to w$.

It is not a classification of all finite-depth accessibility. Pairs whose first possible witness has length at least 3 are outside its scope.

For each generator g, define

$$\mathrm{Exit}_g(v) = \{k : Q_v X_g Q_k \neq 0\},$$

and

$$\mathrm{Entry}_g(w) = \{k : Q_k X_g Q_w \neq 0\}.$$

A length-2 path of color pair (g, h) exists through k if

$$k \in \mathrm{Exit}_g(v) \cap \mathrm{Entry}_h(w).$$

Define

$$T_{g,h,k} = Q_v X_g Q_k X_h Q_w.$$

Then

$$Q_v[X_g, X_h]Q_w = \sum_k (T_{g,h,k} - T_{h,g,k}).$$

Theorem 7.1 (Length-2 R2 Type Classification)**.**
For pairs with length-2 R_1 witnesses, the R_2 behavior falls into four types:

Type	Mechanism	R_2
I	all length-2 paths use a single generator color, so no distinct commutator contributes	0
II	at least one projected commutator block survives	1
III	signed cancellation locus: some $T_{g,h,k}$ are nonzero, but $\sum_k (T_{g,h,k} - T_{h,g,k}) = 0$ for every color pair	0
IV	incidence locus: cross-generator support paths exist, but every relevant product vanishes termwise, $T_{g,h,k} = 0$	0

Proof. If no cross-generator path exists, all depth-1 commutators vanish by antisymmetry, giving Type I. Otherwise cross-generator terms exist. If some projected commutator is nonzero, the pair is Type II. If no projected commutator is nonzero, then either products exist but cancel in each commutator channel, giving Type III, or every relevant product is already zero termwise, giving Type IV. These cases exhaust the length-2 projected commutator expansion.

The separation between Type III and Type IV is structural, not cosmetic. In Type III, the length-2 products are present as matrix products, but the Lie signs place the system on a cancellation locus. In Type IV, the products never reach the signed-sum stage: each candidate product is already killed by an image-kernel alignment. Thus Type III is a relation among surviving path products, while Type IV is a failure of product transversality.

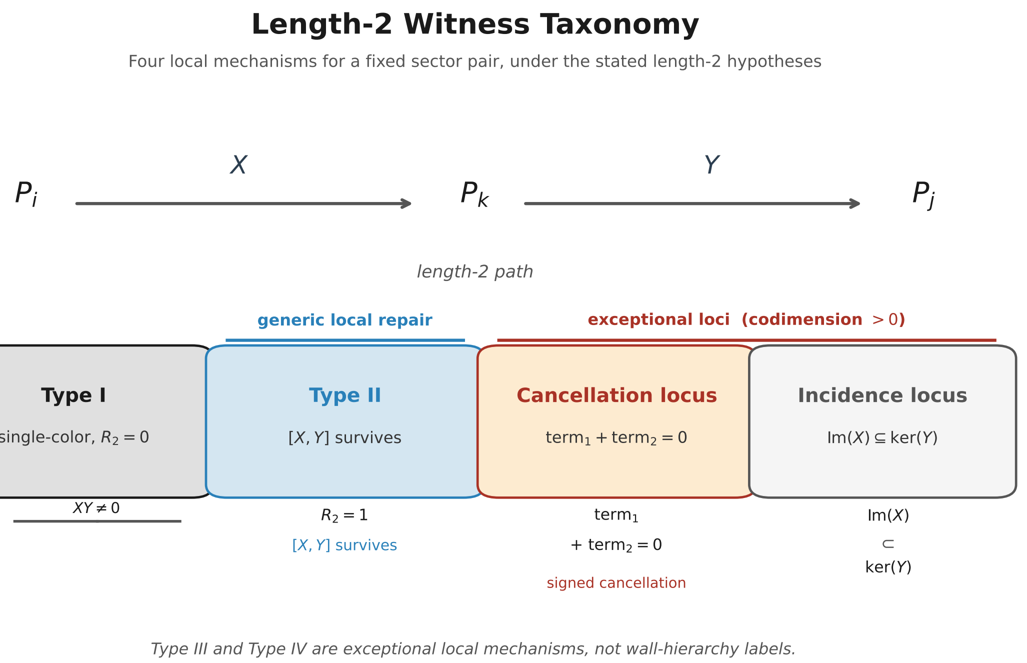

Figure 1. Length-2 witness taxonomy for a fixed sector pair. Type I has only a singleton-color witness and no distinct-generator commutator. Type II is repaired when a projected commutator survives. Type III is a cancellation mechanism: path products exist but cancel in the projected commutator. Type IV is an incidence mechanism: candidate products vanish termwise through image-kernel alignment.

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8 R2 as the First Repair Layer

The repair mechanism is cross-generator commutator survival.

In the complement pattern, take two generators a, b with $u_a = 2$ and $u_b = 3$. Since the generators are skew-Hermitian, a nonzero block in one direction is equivalent to a nonzero adjoint block in the reverse direction. The projected commutator bridge is

$$Q_2[X_a, X_b]Q_3 = Q_2X_aQ_0X_bQ_3 - Q_2X_bQ_0X_aQ_3.$$

The second term is structurally absent because $2 \notin A_b$. Thus

$$Q_2[X_a, X_b]Q_3 = Q_2X_aQ_0X_bQ_3.$$

If this product survives, R_2 creates the bridge $u_a \to u_b$ and destroys the isolated-cycle obstruction.

Lemma 8.1 (Incomparability)**.**

Suppose

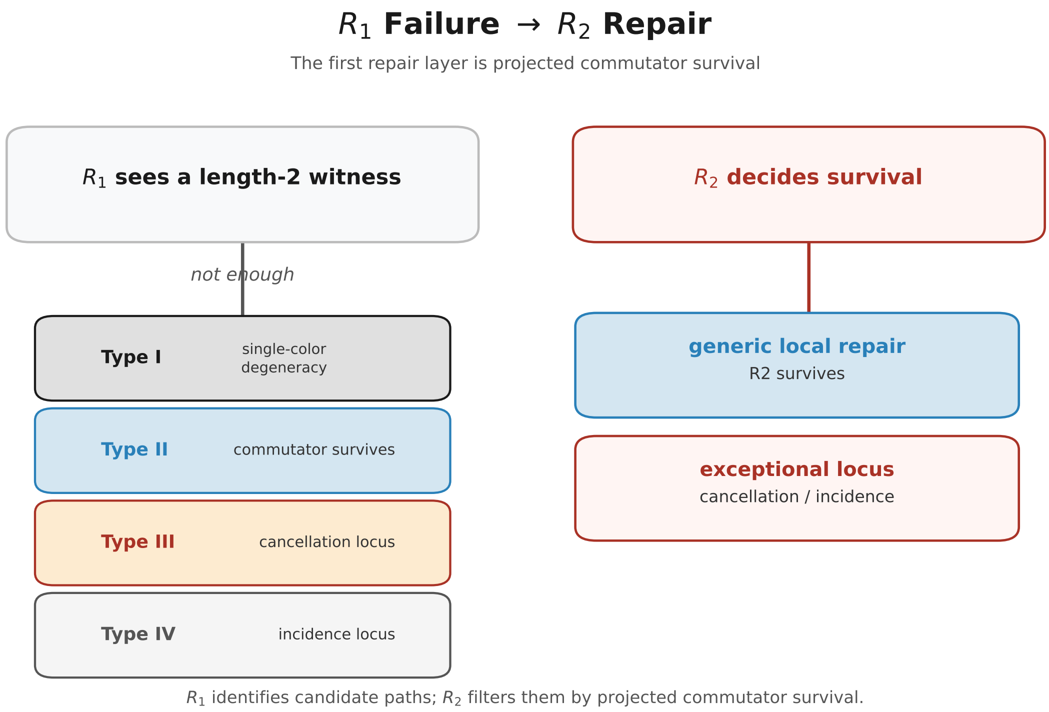

Figure 2. Local repair logic for fixed sectors. The R_1 layer detects candidate length-2 paths but cannot decide whether the projected products survive in a Lie bracket. The R_2 layer records projected commutator survival: in the generic matrix model under the stated non-cancellation and non-incidence hypotheses, repair occurs at the R_2 layer, while Type III and Type IV pairs mark cancellation or incidence loci.

$$A_g \cap B_g = \emptyset, \qquad A_g \cap B_h \neq \emptyset \quad (g \neq h).$$

Then for $g \neq h$, neither $A_g \subseteq A_h$ nor $A_h \subseteq A_g$ is forced in general, but at least $A_g \nsubseteq A_h$ holds for every ordered pair $g \neq h$ satisfying the displayed cross condition.

Proof. *If $A_g \subseteq A_h$, then any vertex in $A_g \cap B_h$ also lies in $A_h \cap B_h$, contradicting $A_h \cap B_h = \emptyset$. Since $A_g \cap B_h$ is nonempty by assumption, this proves $A_g \nsubseteq A_h$.*

For the next lemma we work in the centered support/scalar obstruction model: the relevant length-2 bridge products between the selected sectors pass through the distinguished sector 0, and all other intermediate sector products for the selected block are zero by support. Without this support restriction, other intermediate sector summands may contribute to the projected commutator and the conclusion below is only a candidate mechanism.

Under the local support-obstruction model considered here, only one candidate bridge contributes.

Lemma 8.2 (Single-Term Bridge Candidate)**.**

Under the hypotheses of Lemma 8.1, choose

$$v \in A_g \setminus A_h, \qquad w \in A_h \cap B_g.$$

Then, in the centered support/scalar obstruction model, the commutator block

$$Q_v[X_g, X_h]Q_w$$

has a single surviving support term:

$$Q_v X_g Q_0 X_h Q_w.$$

Proof. *Since $v \in A_g$, the block $Q_v X_g Q_0$ is nonzero. Since $w \in A_h$, the block $Q_w X_h Q_0$ is nonzero; by skew-Hermitian symmetry, the reverse block $Q_0 X_h Q_w$ is also nonzero. Since $v \notin A_h$, the opposite block $Q_v X_h Q_0$ is zero. Therefore*

$$Q_v[X_g, X_h]Q_w = Q_v X_g Q_0 X_h Q_w - 0.$$

At the support level under the centered-model restriction, this is a single-term commutator bridge.

Proposition 8.3 (Support-Level R2 Repair)**.**

In the centered scalar/support obstruction model, R_2 destroys every valid length-2 R_1 bridge obstruction satisfying the hypotheses above.

Proof. *Lemma 8.2 produces a single-term commutator bridge under the centered support restriction. In the scalar/support model, a product of two nonzero scalar factors is nonzero. Therefore the projected commutator survives and creates an R_2 bridge. This bridge breaks the diagonal-empty obstruction.*

Remark 8.4 (Matrix Caveat).

For matrix blocks, nonzero factors do not automatically imply a nonzero product:

$$A \neq 0,\ B \neq 0 \quad \not\Rightarrow \quad AB \neq 0.$$

Thus Proposition 8.3 is a support-level statement in the centered obstruction model. At matrix level, or when additional intermediate sector summands are present, the safe statement is that incomparability identifies a single-term R_2 candidate, while full R_2 records whether the projected commutator actually survives.

This is why R_2 must be relation data, not a value inferred from R_1.

Proposition 8.5 (Conditional Generic Repair and Exceptional Loci)**.**

Fix a binary support pattern G, a list of sector dimensions $\mathbf{d} = (d_i)$, and a finite generator set. Let $\mathcal{R}(G, \mathbf{d})$ be the corresponding matrix-realization variety: forbidden blocks are set equal to zero, and allowed blocks range over the matrix spaces $\mathrm{Mat}_{d_i \times d_j}$. The open support stratum $\mathcal{R}^\circ(G, \mathbf{d}) \subset \mathcal{R}(G, \mathbf{d})$ consists of realizations in which every block allowed by G is nonzero.

Assume that the relevant length-2 candidate products exist and that the projected commutator polynomial is not identically zero on the chosen support stratum. Inside $\mathcal{R}^\circ(G, \mathbf{d})$, Types III and IV are nongeneric algebraic exceptional loci. They are different exceptional mechanisms.

Type III is the cancellation locus cut out by the projected commutator equations

$$\sum_k (T_{g,h,k} - T_{h,g,k}) = 0 \qquad \text{for all relevant } (g,h),$$

with at least one summand nonzero. It is a signed-product relation among existing length-2 paths.

Type IV is the incidence locus where each candidate product is termwise zero. For a single product AB, this is the image-kernel alignment

$$\mathrm{im}(B) \subseteq \ker(A).$$

Thus, under the stated nonzero-polynomial and candidate-product hypotheses, away from these exceptional loci non-Type-I length-2 witnesses survive at R_2. The Type IV condition is naturally an incidence or Schubert-type condition on subspaces [7, 5]. This generic repair statement is local and conditional. It does not prove global accessibility completeness, and it does not apply to pairs whose first possible witness has Hall length at least 3*.*

The proposition is the static mechanism layer beneath the later wall-crossing question. In deformation language, Type III records signed commutator cancellation and Type IV records product incidence. These are not additional names for the Paper VI wall-hierarchy loci Σ_{R_1}, $\Sigma^\circ_{R_2}$, or Σ°_D, and they are not a wall classification scheme. Rather, they are local mechanism strata that may realize pieces of the residual R_2 wall after R_1 has been held fixed.

Remark 8.6 (Algebraic Accessibility Structures).

The Type III/Type IV distinction is the first place where the accessibility calculus becomes algebraic rather than combinatorial. Type III is a relation among existing products: the length-2 paths exist as matrices, but their signed sum lies on a cancellation locus. Type IV is a product-incidence condition: the candidate products vanish termwise because an image lies in a kernel.

Thus the failure and success pattern of

$$(R_1, R_2) \longrightarrow D$$

is controlled by two different algebraic structures. Cancellation can often be broken by adding higher-depth Hall directions; incidence is a nontransversality condition for the block product itself. Later completion theory uses this distinction to explain why, in all tested structured systems with exact (R_1, R_2) signature matches, the depth data D is empirically constant on each signature class, while Type IV remains the natural exceptional boundary.

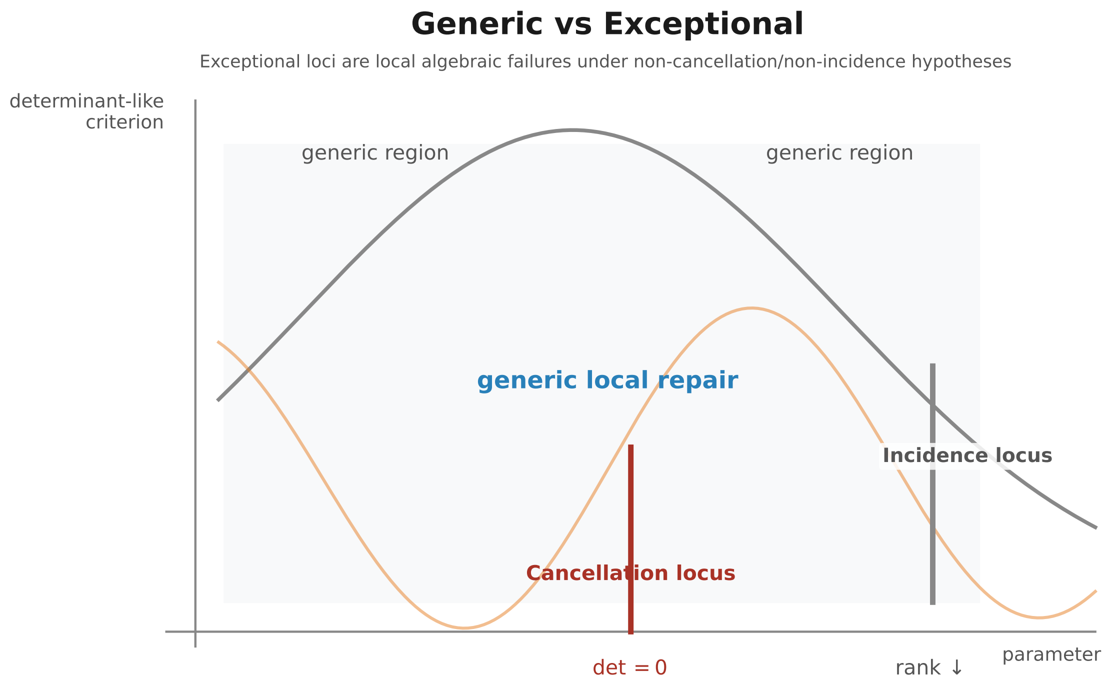

Figure 3. Generic versus exceptional local geometry for length-2 witnesses. In the generic matrix model, under the stated nonzero-polynomial and candidate-product hypotheses, the relevant projected products survive away from algebraic cancellation and incidence loci and the witness is repaired at the R_2 layer. The cancellation locus is represented by a polynomial vanishing condition; the incidence locus is represented by rank drop or image-kernel alignment.

9 Matrix Nondegeneracy

The single-term bridge has matrix form

$$A = Q_v X_g Q_k : \mathbb{C}^{d_k} \to \mathbb{C}^{d_v},$$

$$B = Q_k X_h Q_w : \mathbb{C}^{d_w} \to \mathbb{C}^{d_k}.$$

The product is

$$AB = Q_v X_g Q_k X_h Q_w.$$

Lemma 9.1 (Rank Protection)**.**
If A has full column rank and $B \neq 0$, then $AB \neq 0$. Dually, if B has full row rank and $A \neq 0$, then $AB \neq 0$.

Proof. *If A has full column rank, then $\ker A = \{0\}$. If $AB = 0$, then every vector in the image of B lies in $\ker A$, so $\operatorname{im} B = 0$ and hence $B = 0$, contradiction. The dual statement follows by replacing full column rank with full row rank and using $\operatorname{im} B = \mathbb{C}^{d_k}$.*

This is the elementary form of the rank obstruction underlying the usual matrix-rank nondegeneracy tests, compatible with the Sylvester-rank viewpoint in standard matrix analysis [8].

The useful dimension condition is

$$d_k \leq \min(d_v, d_w).$$

Under generic full-rank behavior, this permits A to have full column rank and B to have full row rank.

9.1 S4 Three-Generator Verification

The S4-3gen-B matrix nondegeneracy check found:

Quantity	Result
single-term bridge products	48
$AB = 0$ cases	0
bridges satisfying $d_k \leq \min(d_v, d_w)$	48/48
perturbation trials	100/100 with no $AB = 0$

This does not prove a universal matrix theorem. It identifies the right matrix-level certificate for Proposition 8.3: single-term bridge products need rank protection, direct verification, or inclusion in R_2 as relation data.

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10 Jacobi and Higher Layers

The free Lie algebra has two primitive relation sources relevant here:

1. antisymmetry of the bracket, already visible in R_2;
2. the Jacobi identity, first visible at the next nested layer.

We use only these standard Lie-algebraic relations and their projected consequences [9, 11].

Earlier evidence suggested Jacobi might always be latent. Later sparse random graph searches corrected this: Jacobi latency is a density phenomenon.

Sparse artificial sector graphs can exhibit non-latent R_3 cases. In medium, dense, and representation-like tests, R_2 usually connects first, making Jacobi redundant for the first depth matrix.

The current boundary is:

Statement	Status
universal Jacobi latency for arbitrary quivers	false
Jacobi latency in sparse random graphs	false at low frequency
Jacobi latency in dense or represented systems	strong evidence
arbitrary-quiver $N = 2$ completeness	false or unsupported
represented/dense $N = 2$ completeness	program-level conjecture

Therefore Paper V should not claim a universal $N = 2$ theorem. The correct theorem program is domain-sensitive.

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11 Boundary of the Static Theory

The stable results of this paper are Theorem 7.1, Proposition 8.3, and Proposition 8.5: a local classification of length-2 R_1 witnesses, a centered support-level repair theorem, and a conditional generic exceptional-locus statement. These results deliberately stop short of a full depth theorem.

11.1 Matrix-Level Local R2 Repair

For a representation-derived sectorized system, a valid length-2 R_1 bridge obstruction is repaired or certified by full projected R_2 when each relevant bridge product has a concrete nondegeneracy certificate: rank protection, direct nonzero-product verification, or explicit projected R_2 survival.

This is proved at centered support/scalar level by the single-term bridge mechanism. The matrix-level upgrade requires rank protection, direct nondegeneracy verification, or accepting R_2 as the relation oracle.

11.2 Length-2 Witness Reduction

Every finite-depth accessible pair with $R_1 = 0$ in the represented systems of interest admits a length-2 R_1 witness.

This is the global Layer C reduction. It is open.

11.3 Conditional Local Completeness

If the matrix-level local repair program and the length-2 witness reduction both hold for a class of sectorized systems, then one obtains the candidate conditional theorem

$$(R_1, R_2) \to D$$

for that class.

This is the desired Accessibility Completeness theorem for that class. It is not claimed here as proved, and it is not a theorem of the present paper.

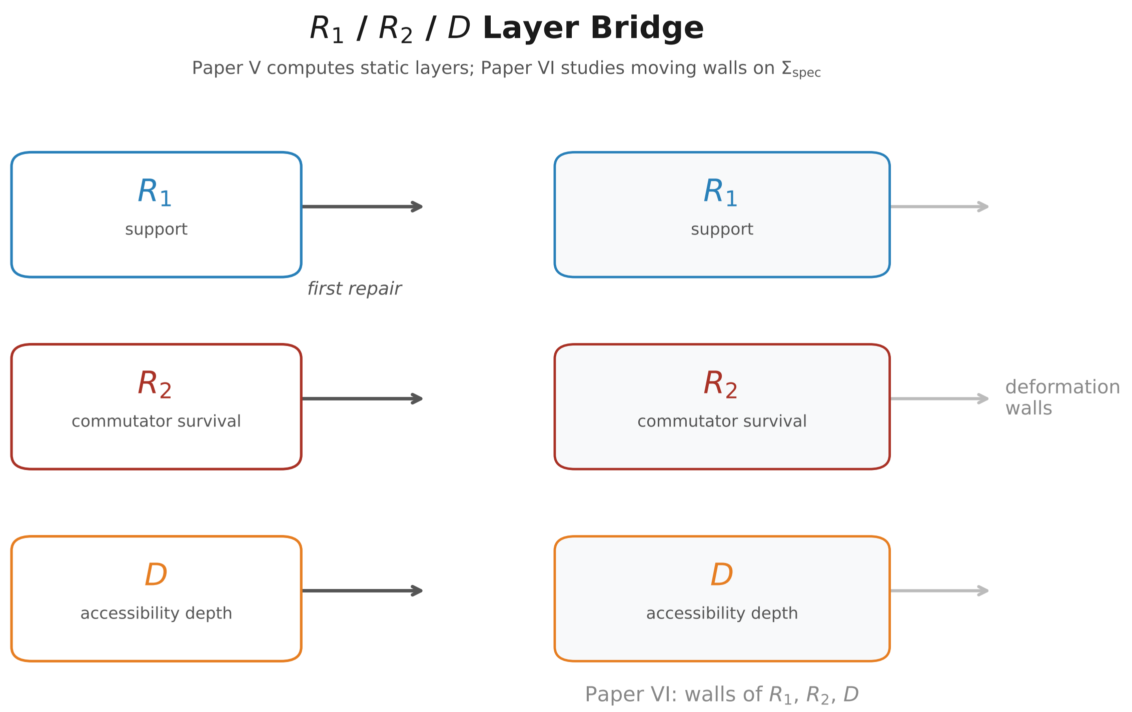

Figure 4. Static-to-dynamic bridge. Paper V studies fixed-system data R_1, R_2, and D: support, projected commutator survival, and first accessibility depth. Paper VI moves the same observable shadows over a parameter space and studies the walls where their rank/support profiles jump.

11.4 Relation to the Deformation Stratification

This program-level status is compatible with the accessibility stratification developed in Paper VI. Paper VI is local in moduli: on normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, it studies where the discrete observables $R_1(w)$, $R_2(w)$, and $D(w)$ remain locally constant and where they jump. Its wall hierarchy is an observable-level stratification,

$$\widehat{\Sigma}_{R_1} \subseteq \widehat{\Sigma}_{R_2} \subseteq \widehat{\Sigma}_D.$$

Paper V asks a different question. It fixes a sectorized system and a sector pair, then classifies the length-2 mechanisms by which R_1 may fail and R_2 may repair or fail to repair it. Thus Paper VI supplies a local wall hierarchy for moving systems, while Paper V supplies mechanism types for the static fibers over those walls. Neither statement implies the global represented/dense completeness theorem $(R_1, R_2) \to D$ by itself.

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12 Relation to Paper III

Paper III's T7 result shows a separation between composition-generated accessibility and Lie-generated accessibility in the Rubik cubie representation. It is a witness that accessibility depends on sector projectors and composition, not only on the Lie algebra. The terminology of Lie-generated accessibility follows the usual Lie-algebraic control-theoretic lineage [10].

Paper V asks a different question: at what first depth does the projected Lie algebra see a sector pair?

Thus Paper III supplies the seed example and motivation. Paper V supplies the minimal-data problem for first-depth emergence.

The two are related but not identical:

Paper III	Paper V
Rubik-centered separation example	general sectorized observable framework
Lie vs composition	first Lie-depth inside the Lie filtration
T7 morphisms	R_1, R_2, weighted Hall paths
cross-block composition gap	projected commutator survival and cancellation

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13 Computational Support

The reproducibility suite records the finite examples used to test the local repair calculus. It is not part of the mathematical definitions; it keeps each computation attached to the manuscript statement it illustrates or audits.

Label	Manuscript interface	Role
S4 depth	Proposition 5.1; Theorem 7.1	S4-3gen-B sectorized example with signature $(10, 2, 2, 76)$ and depth-2 behavior
PCM	Proposition 5.1; Theorem 7.1 Type III	binary-support counterexample: predicted depth 1, observed depth 2 by projected commutator cancellation
Complement	Proposition 8.3	scalar/support complement obstruction and nonzero single-term R_2 bridge repair
Non-complement	Proposition 8.3 boundary	finite enumeration showing complement is not the only R_1 obstruction family
Matrix	Lemma 9.1; Proposition 8.5 Type IV boundary	S4-3gen-B bridge audit: 48/48 nonzero products and rank-protected bridges

The repository groups these audits under the Paper V experiment suite: S4 depth, path-commutator cancellation, complement and non-complement obstruction enumeration, and matrix nondegeneracy.

The mapping is intentionally tight. Proposition 5.1 is witnessed by the explicit failure of binary support to determine D. Theorem 7.1 is proved by the local length-2 expansion; the audits instantiate the Type III case and check the finite examples used in the classification. Proposition 8.3 is illustrated at the support/scalar level by the single-term bridge computations. Lemma 9.1 and the Type IV side of Proposition 8.5 are audited by the matrix nondegeneracy computation, which shows that the expected image-kernel incidence failure does not occur in the S4-3gen-B bridge family.

Additional deformation and Type IV search audits are reserved for the later deformation theory. They motivate the representation-derived exceptional-locus question, but they are not used as claim support in this paper. In the language of Paper VI, the current suite exhibits a Type III mechanism that may occur inside a residual $\Sigma^{\circ}_{R_2}$ wall, together with a tested absence of Type IV in one structured matrix-block bridge family; it does not classify all represented exceptional loci.

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14 Long-Term Program

The preceding sections close the local length-2 repair calculus. They do not close the global accessibility-completeness problem. The remaining program splits into four independent questions.

14.1 Monochromatic Closure

Type I pairs have length-2 paths, but no distinct-generator commutator contributes. The local classification proves only $R_2 = 0$. It does not prove that the pair is frozen in the full Hall filtration.

The closure problem asks whether a Type I pair can become visible through a longer Hall word, or whether it is closed under the full Lie filtration.

This is the first boundary of the local calculus.

14.2 Length-2 Reduction

The local theory assumes the existence of a length-2 R_1 witness. The global reduction asks whether every finite-depth accessible pair with $R_1 = 0$ admits a length-2 R_1 witness.

If the answer is negative, then some accessibility phenomena begin only at Hall length at least 3, outside the scope of this paper. If the answer is positive for a represented class of systems, then the local R_2 calculus becomes a candidate depth detector for that class.

14.3 Representation-Derived Exceptional Loci

The local classification identifies two nongeneric depth-1 loci. Type III is the signed cancellation locus: length-2 products survive, but the projected commutator sum vanishes. Type IV is the incidence locus: candidate products vanish termwise through $AB = 0$.

The representation-derived question is to determine which of these exceptional loci are actually realized by structured systems. The current evidence is asymmetric: Type III appears in finite-group examples such as S4-3gen-B, while Type IV is absent in the tested Wedderburn-coordinate regular-representation sectorizations and is controlled by rank/nondegeneracy conditions in the S4 bridge audit.

This is the point where Paper V feeds Paper VI. Type III and Type IV are not Paper VI wall-hierarchy labels, and they should not be read as wall classification labels. They are mechanism-level strata inside the static fiber of the deformation problem: Type III is the cancellation component of residual R_2 failure, and Type IV is the incidence component. The observable-level walls Σ_{R_1}, $\Sigma^{\circ}_{R_2}$, and Σ°_{D} remain the Paper VI classification axis.

14.4 Conditional Accessibility Completeness

Only after the previous questions are settled for a chosen class of sectorized systems should one promote the global statement

$$(R_1, R_2) \to D$$

from a program to a theorem.

The conditional route combines local length-2 R_2 repair, matrix nondegeneracy or direct R_2 survival, length-2 witness reduction, and control of represented exceptional loci. Together, these ingredients would yield a candidate class-level completeness theorem.

Thus $(R_1, R_2) \to D$ is not the main theorem of Paper V. It is the long-term completeness program opened by the repair calculus.

Paper VII [2] takes up precisely this program. There, the Type IV mechanism is reinterpreted as the incidence variety

$$I = \{(A, B) : AB = 0,\ A \neq 0,\ B \neq 0\},$$

whose fixed-rank strata have quadratic codimension growth. In that later completion theory, Type IV is no longer only a local exceptional label: it is the high-codimension algebraic boundary outside which the Generic Completion Principle is formulated.

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15 Conclusion

The accessibility problem has moved from binary graph theory to weighted Hall path algebra. The binary graph (G, χ, Λ) records generator support, but it does not record projected product cancellation. Therefore it cannot determine first Lie-accessibility depth.

The first repair layer is R_2, the survival of projected commutators. At support level, valid R_1 bridge obstructions force single-term R_2 bridge candidates by incomparability. These bridges destroy the obstruction. At matrix level, their survival is controlled by rank/nondegeneracy or must be recorded directly as relation data.

The current stable statement has five parts: R_1 fails structurally; R_2 is the first repair layer; length-2 accessibility witnesses admit a local Type I–IV calculus with distinct Type III and Type IV exceptional mechanisms; accessibility depth is a valuation of weighted Hall coefficients; and full $(R_1, R_2) \to D$ completeness remains a long-term program.

This is the natural post-T7 theory. Paper III shows that accessibility separates. Paper V identifies the first repair calculus for length-2 witnesses, explains why binary support must be replaced by weighted Hall path data before any global depth theorem can be attempted, and isolates local mechanisms that later deformation theory can see inside accessibility walls.

Bibliographic Notes

Program lineage. Paper V continues the accessibility line opened by Paper III. Paper III supplies the Lie-generated versus composition-generated separation in the Rubik cubie representation [1]; the CCS records the canonical sector and transport data used by the trilogy [4]. Paper V extracts the static local repair calculus for fixed sectorized systems and prepares the $R_1/R_2/D$ language used later by Papers VI and VII [3, 2].

External background. The weighted Hall path algebra uses Hall's construction of Hall bases and the modern free-Lie-algebra treatment of Hall words and Lie monomials [6, 11]. The Lie-depth viewpoint is aligned with geometric control theory and standard Lie-algebra representation conventions [10, 9]. Rank protection uses elementary matrix-analysis principles [8]. Representation-derived examples use finite-group representation theory [12]. The incidence-locus language for Type IV uses the standard incidence/Schubert viewpoint from algebraic geometry [7, 5].

Computational provenance. The manuscript-level support consists of the S4 R_1/R_2 depth example, the path-commutator cancellation audit, complement and non-complement obstruction enumerations, and the matrix-nondegeneracy check. The repository stores the corresponding support files, figures, and canonical tables.

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